

Case Study: Employee Payroll and Performance Analysis Using Java 8

1. Case Study Overview

ABC Technologies is a mid-sized software company with employees working across multiple departments such as:

- Information Technology
- Human Resources
- Finance
- Sales
- Operations
- Administration

The HR department maintains employee information manually using spreadsheets. Every month, the HR and Finance teams prepare different employee reports, calculate salaries, identify promotion-eligible employees, calculate bonuses and analyse department-wise salary expenses.

As the number of employees increases, preparing these reports manually becomes time-consuming and error-prone.

The company has decided to develop an **Employee Payroll and Performance Analysis System** using Java 8 features.

2. Business Problem

The existing employee management process has the following problems:

- Employee records are processed manually.
- Salary reports take significant time to prepare.
- HR must manually identify high-performing employees.
- Bonus calculations may contain errors.
- Department-wise reports are difficult to generate.
- Finding employees based on salary, department or experience takes time.
- Missing employee records are not handled properly.
- Payroll processing blocks other HR activities.
- Employee joining dates and experience calculations are performed manually.

The organisation needs an automated Java application to process employee information efficiently.

3. Project Objective

The objective of this project is to create an application that uses Java 8 features to perform employee payroll and performance analysis.

The system should allow the organisation to:

- Store and manage employee information.
 - Filter employees based on different conditions.
 - Sort employee records.
 - calculate salary-related information.
 - Analyse employee performance.
 - Generate department-wise reports.
 - Handle missing employee information safely.
 - Process payroll-related operations asynchronously.
 - Calculate employee experience using the Date and Time API.
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4. Employee Information

Each employee record should contain the following information:

Field	Description
Employee ID	Unique identification number
Employee name	Full name of the employee
Department	Department in which the employee works
Designation	Employee's job title
Salary	Monthly salary
Joining date	Date on which the employee joined
Performance rating	Rating between 1 and 5
Employment status	Active or inactive
Email address	Official email address
Manager name	Reporting manager
Location	Employee's work location

5. Departments

The system should contain employees from departments such as:

- IT
- HR
- Finance
- Sales
- Operations
- Administration

Each department should have multiple employees with different:

- Salaries
 - Joining dates
 - Performance ratings
 - Designations
 - Employment statuses
-

6. Functional Requirements

6.1 Display All Employees

The system should display all available employee records.

The output should contain:

- Employee ID
- Employee name
- Department
- Designation
- Salary
- Joining date
- Performance rating
- Employment status

Java 8 concepts

- Consumer
 - `forEach()`
 - Method reference
-

6.2 Find Active Employees

The system should identify employees whose employment status is active.

Inactive employees should not be included in active payroll processing.

Java 8 concepts

- Predicate
 - Lambda expression
 - Stream filtering
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6.3 Find Employees by Department

The HR user should be able to provide a department name and retrieve all employees working in that department.

Example departments:

- IT
- HR
- Finance
- Sales

Java 8 concepts

- Lambda expression
 - Predicate
 - `filter()`
-

6.4 Find High-Salary Employees

The system should identify employees earning more than a specified salary amount.

For example:

- Employees earning more than ₹50,000
- Employees earning more than ₹75,000
- Employees earning more than ₹1,00,000

The salary limit should not be fixed permanently. It should be provided as an input.

Java 8 concepts

- Predicate
 - Lambda expression
 - Stream API
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6.5 Sort Employees by Salary

The system should sort employees based on salary.

It should support:

- Lowest salary to highest salary
- Highest salary to lowest salary

Java 8 concepts

- Comparator
 - Lambda expression
 - Method reference
 - `sorted()`
-

6.6 Sort Employees by Name

The application should arrange employees alphabetically based on their names.

It should support:

- Ascending alphabetical order
- Descending alphabetical order

Java 8 concepts

- Comparator
 - Method reference
 - Stream sorting
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6.7 Sort Employees by Joining Date

The system should sort employees based on their joining dates.

This will help HR identify:

- Recently joined employees
- Longest-serving employees

Java 8 concepts

- Comparator
 - Date and Time API
 - Method reference
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6.8 Convert Employee Names to Uppercase

The application should generate a list containing employee names in uppercase.

The original employee records should remain unchanged.

Java 8 concepts

- Function
 - `map()`
 - Lambda expression
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6.9 Generate Employee Name List

The application should extract only employee names from complete employee records.

The resulting collection should contain names instead of full employee objects.

Java 8 concepts

- Function
 - `map()`
 - Stream transformation
-

6.10 Calculate Total Salary Expense

The system should calculate the total monthly salary expense of all employees.

It should also calculate the total salary expense of active employees separately.

Java 8 concepts

- `reduce()`
 - `mapToDouble()`
 - `sum()`
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6.11 Calculate Average Salary

The application should calculate:

- Overall average salary
- Average salary of active employees
- Average salary of each department

Java 8 concepts

- Stream API
 - `average()`
 - `Collectors.averagingDouble()`
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6.12 Find Highest-Paid Employee

The system should identify the employee receiving the highest salary.

The result should display:

- Employee name
- Department
- Designation
- Salary

Java 8 concepts

- Comparator
 - `max()`
 - Optional
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6.13 Find Lowest-Paid Employee

The system should identify the employee receiving the lowest salary.

The result should display basic employee and salary information.

Java 8 concepts

- Comparator
 - `min()`
 - Optional
-

6.14 Find Top Three Highest-Paid Employees

The system should generate a list of the top three highest-paid employees.

The results should be arranged from highest to lowest salary.

Java 8 concepts

- Sorting
 - `limit()`
 - Stream API
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6.15 Group Employees by Department

The system should group employees based on their departments.

Example result:

- IT department employees
- HR department employees
- Finance department employees
- Sales department employees

Java 8 concepts

- `Collectors.groupingBy()`
 - Method reference
 - Map processing
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6.16 Count Employees in Each Department

The system should calculate the number of employees working in every department.

Example:

Department	Employee count
IT	12
HR	5
Finance	7
Sales	10

Java 8 concepts

- `groupingBy()`
 - `counting()`
 - Collectors
-

6.17 Calculate Department-Wise Salary Expense

The application should calculate the total salary expense for each department.

This report will help management understand departmental payroll costs.

Java 8 concepts

- `groupingBy()`
 - `summingDouble()`
 - Collectors
-

6.18 Partition Active and Inactive Employees

The application should divide employees into two groups:

- Active employees
- Inactive employees

Java 8 concepts

- `partitioningBy()`
 - Predicate
 - Collectors
-

6.19 Calculate Employee Experience

The system should calculate the employment experience of each employee using the joining date.

Experience should be shown in:

- Years
- Months

Java 8 concepts

- `LocalDate`
 - `Period`
 - Java 8 Date and Time API
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6.20 Find Longest-Serving Employee

The system should identify the employee who joined the organisation earliest.

The result should display:

- Employee name
- Joining date
- Total experience
- Department

Java 8 concepts

- Date comparison
 - Comparator
 - `min()`
 - Optional
-

6.21 Find Employees Who Completed Five Years

The system should identify employees who have completed at least five years in the organisation.

Only active employees should be considered.

Java 8 concepts

- Date and Time API
 - Predicate
 - Stream filtering
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6.22 Find Employees Who Joined in a Particular Year

The HR user should be able to provide a year and retrieve employees who joined during that year.

Example:

- Employees who joined in 2021
- Employees who joined in 2024

Java 8 concepts

- `LocalDate`
 - Lambda expression
 - Filtering
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6.23 Find Employees with Work Anniversary in the Current Month

The system should identify employees whose joining month matches the current month.

This report can be used to send work anniversary messages.

Java 8 concepts

- `LocalDate`
 - Date comparison
 - Stream filtering
-

6.24 Identify Promotion-Eligible Employees

An employee should be considered eligible for promotion when all the following conditions are satisfied:

- Employee is active.
- Performance rating is at least 4.0.
- Employee has completed at least three years.
- Employee has not received a recent promotion.

The application should generate a promotion eligibility report.

Java 8 concepts

- Multiple predicates
- Predicate chaining
- Date and Time API
- Stream API

6.25 Calculate Employee Bonus

The application should calculate bonuses according to employee performance.

Bonus rules

Performance rating	Bonus percentage
4.5 to 5.0	15%
4.0 to 4.49	10%
3.5 to 3.99	5%
Below 3.5	No bonus

The bonus should be calculated using a custom functional interface.

Java 8 concepts

- Functional interface
- Lambda expression
- Function-based calculation

6.26 Search Employee by ID

The system should search for an employee using the employee ID.

Possible results:

- Employee record found
- Employee record not found

The system should not return or directly process a null value.

Java 8 concepts

- Optional
- `findFirst()`
- `orElse()`
- `orElseGet()`
- `ifPresent()`

6.27 Search Employee by Email

The HR user should be able to search for an employee using the official email address.

Email comparison should be case-insensitive.

Java 8 concepts

- Stream filtering
 - Optional
 - Lambda expression
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6.28 Generate Employee IDs

The system should generate new employee IDs for newly added employees.

The ID generation logic should be implemented using a supplier.

Java 8 concepts

- Supplier
 - Lambda expression
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6.29 Validate Employee Salary

The system should verify whether an employee salary is valid.

A salary should be considered valid when:

- Salary is greater than zero.
- Salary satisfies the minimum salary policy.
- Salary does not exceed the defined designation limit.

The validation logic can be provided through a static method in an interface.

Java 8 concepts

- Static method in interface
 - Predicate
 - Validation logic
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6.30 Display HR Policy

The employee service interface should provide a default method to display the organisation's HR or promotion policy.

Implementing classes should be able to use the default implementation or override it when required.

Java 8 concepts

- Default method in interface
 - Interface enhancement
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6.31 Process Payroll Asynchronously

Payroll processing may involve:

- Calculating salary totals
- Calculating bonuses
- Generating salary reports
- Sending payroll completion notifications

These activities should run asynchronously without blocking the main application process.

Java 8 concepts

- `CompletableFuture`
 - Asynchronous processing
 - Task chaining
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6.32 Generate Department Reports Asynchronously

The system should generate reports for different departments independently.

For example:

- IT salary report
- HR salary report
- Finance salary report
- Sales salary report

Each report should be processed as a separate asynchronous task.

Java 8 concepts

- Multiple `CompletableFuture` tasks
 - Combining asynchronous results
 - Parallel processing
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7. Recommended Project Architecture

The project should follow a layered architecture.

7.1 Model Layer

The model layer should contain the employee data structure.

Suggested class:

- `Employee`

Responsibilities:

- Store employee information.
 - Provide constructors.
 - Provide getter and setter methods.
 - Provide a readable employee representation.
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7.2 Repository Layer

The repository layer should manage employee data.

Suggested components:

- `EmployeeRepository` interface
- `EmployeeRepository` implementation

Responsibilities:

- Add employee records.
- Retrieve all employees.
- Search employees by ID.
- Search employees by email.
- Update employee details.
- Remove employee records.
- Store sample employee data.

The data may initially be stored using an in-memory collection.

7.3 Service Layer

The service layer should contain business rules and Java 8 processing.

Suggested components:

- EmployeeService interface
- EmployeeService implementation
- BonusCalculator functional interface

Responsibilities:

- Filter employees.
 - Sort employee records.
 - Calculate payroll.
 - Calculate bonuses.
 - Determine promotion eligibility.
 - Group employees by department.
 - Calculate employee experience.
 - Generate reports.
 - Perform asynchronous operations.
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7.4 Utility Layer

The utility layer should contain reusable supporting functions.

Suggested utilities:

- Employee ID generator
 - Date calculation utility
 - Salary validator
 - Report formatter
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7.5 Application Layer

The application layer should act as the entry point.

Responsibilities:

- Create repository and service objects.

- Load sample employee records.
 - Call different operations.
 - Display reports.
 - Handle user input.
 - Display suitable error messages.
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8. Step-by-Step Implementation Guide

Step 1: Create the Project

Create a Java project using Java 8 or above.

Create the following packages:

- Model
 - Repository
 - Service
 - Utility
 - Application
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Step 2: Create the Employee Model

Create an employee model containing all required fields.

Include:

- Employee ID
- Name
- Department
- Designation
- Salary
- Joining date
- Performance rating
- Status
- Email
- Manager
- Location

Add suitable constructors and accessor methods.

Step 3: Prepare Sample Employee Records

Create at least 15 employee records.

Ensure the records contain:

- Multiple departments
- Different salaries
- Different joining years
- Different ratings
- Active and inactive employees
- Multiple locations

This variety is required to test filtering, sorting and grouping.

Step 4: Create the Repository Layer

Create a repository contract to define employee data operations.

The repository should support:

- Adding an employee
- Retrieving all employees
- Searching by ID
- Searching by email
- Updating an employee
- Removing an employee

Create an implementation using a collection.

Step 5: Create a Functional Interface for Bonus Calculation

Create a functional interface containing one abstract method.

The method should accept:

- Employee salary
- Performance rating

It should return:

- Calculated bonus amount

Implement the bonus rules using a lambda expression.

Step 6: Create the Employee Service Interface

Define employee processing operations in the service interface.

Include operations for:

- Filtering employees
- Sorting employees
- Calculating salary
- Grouping employees
- Calculating experience
- Finding promotion eligibility

Add:

- A default method for displaying HR policy
- A static method for salary validation

Step 7: Implement the Service Layer

Create the service implementation.

Use Stream API operations instead of traditional external loops wherever possible.

Implement the required:

- Filters
- Transformations
- Sorting
- Aggregations
- Grouping
- Partitioning
- Searching

Step 8: Implement Predicate-Based Filters

Create reusable predicates for:

- Active employees
- High-salary employees
- High-performing employees
- Department-based employees

- Promotion-eligible employees
- Employees with five years of experience

Combine predicates wherever multiple conditions are required.

Step 9: Implement Function-Based Transformations

Use functions to transform employee records into:

- Employee names
- Uppercase employee names
- Employee salaries
- Department names
- Employee summary strings

Ensure that transformation operations do not modify the original collection.

Step 10: Implement Consumer-Based Display

Use consumers for displaying:

- Complete employee details
- Employee names
- Payroll summaries
- Department reports
- Bonus reports

Use method references where an existing method matches the required operation.

Step 11: Implement Supplier-Based ID Generation

Create an employee ID supplier.

Ensure that:

- Generated IDs follow an agreed format.
 - Duplicate IDs are not assigned.
 - The generated ID can be used while adding a new employee.
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Step 12: Implement Sorting Operations

Create sorting operations for:

- Employee name
- Salary
- Joining date
- Performance rating
- Department

Provide both ascending and descending options where applicable.

Step 13: Implement Salary Calculations

Calculate:

- Total salary of all employees
 - Total salary of active employees
 - Average salary
 - Highest salary
 - Lowest salary
 - Department-wise total salary
 - Department-wise average salary
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Step 14: Implement Department Grouping

Group employees by department.

For each department, display:

- Employee count
 - Employee names
 - Total salary
 - Average salary
 - Highest-paid employee
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Step 15: Implement Active and Inactive Partitioning

Separate the complete employee collection into:

- Active employee group
- Inactive employee group

Display the count and employee records in each group.

Step 16: Implement Date-Based Operations

Using Java 8 Date and Time API, calculate:

- Employee experience
 - Employees who completed five years
 - Employees who joined in a particular year
 - Work anniversaries in the current month
 - Longest-serving employee
 - Most recently joined employee
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Step 17: Implement Promotion Eligibility

Create business rules for promotion eligibility.

Check:

- Active status
- Minimum rating
- Minimum experience
- Previous promotion details

Generate a separate report for eligible and non-eligible employees.

Step 18: Implement Optional Handling

Use Optional while searching for:

- Employee by ID
- Employee by email
- Highest-paid employee
- Lowest-paid employee
- Longest-serving employee

Display a proper message when no result is available.

Avoid returning null from search operations.

Step 19: Implement Bonus Calculation

Calculate bonuses using the custom functional interface.

Generate a bonus report containing:

- Employee ID
 - Employee name
 - Salary
 - Rating
 - Bonus percentage
 - Bonus amount
 - Total amount after bonus
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Step 20: Implement Payroll Processing

Calculate payroll only for active employees.

Payroll should include:

- Basic salary
 - Bonus
 - Total payable amount
 - Department-wise payroll
 - Overall payroll expense
-

Step 21: Implement Asynchronous Payroll

Use asynchronous processing for:

- Overall payroll calculation
- Bonus calculation
- Department report generation
- Payroll completion notification

The main application should continue executing while these tasks run.

Step 22: Handle Exceptions

Add suitable validation and exception handling for:

- Invalid employee ID

- Negative salary
 - Invalid rating
 - Missing department
 - Invalid joining date
 - Duplicate employee ID
 - Employee not found
 - Empty employee collection
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Step 23: Display Final Reports

Generate the following reports:

1. Complete employee report
 2. Active employee report
 3. Inactive employee report
 4. Department-wise employee report
 5. High-salary employee report
 6. Promotion eligibility report
 7. Employee experience report
 8. Bonus report
 9. Payroll report
 10. Work anniversary report
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9. Business Rules

9.1 Employee ID Rule

- Employee ID must be unique.
- Employee ID must not be empty.
- Automatically generated IDs should not duplicate existing IDs.

9.2 Salary Rule

- Salary must be greater than zero.
- Salary must meet the company's minimum salary policy.
- Salary should be validated based on designation.

9.3 Performance Rating Rule

- Rating must be between 1.0 and 5.0.
- Employees with a rating below 3.5 are not eligible for a bonus.
- Employees with a rating of 4.0 or above may be considered for promotion.

9.4 Promotion Rule

An employee is eligible when:

- Employment status is active.
- Rating is at least 4.0.
- Experience is at least three years.
- There is no disciplinary issue.
- The employee has not recently received a promotion.

9.5 Payroll Rule

- Only active employees are included in payroll.
 - Inactive employees should appear in a separate report.
 - Bonus should be added only after performance validation.
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10. Java 8 Feature Mapping

Java 8 feature	Application in the case study
Lambda expression	Filtering, sorting, bonus calculation
Functional interface	Employee bonus calculation
Predicate	Salary, department, status and eligibility checks
Function	Employee data transformation
Consumer	Displaying employee and report data
Supplier	Employee ID generation
Method reference	Displaying and extracting employee information
Stream API	Processing employee collections
<code>filter()</code>	Selecting matching employees
<code>map()</code>	Transforming employee data
<code>sorted()</code>	Ordering employee records
<code>reduce()</code>	Calculating salary totals
<code>groupingBy()</code>	Grouping employees by department
<code>partitioningBy()</code>	Separating active and inactive employees
Optional	Handling missing employee records

Java 8 feature	Application in the case study
Date and Time API	Experience and anniversary calculations
Default interface method	HR policy implementation
Static interface method	Salary validation
CompletableFuture	Asynchronous payroll and report generation

11. Expected Reports

11.1 Employee Summary Report

The report should contain:

- Total number of employees
 - Number of active employees
 - Number of inactive employees
 - Total salary expense
 - Average salary
 - Highest salary
 - Lowest salary
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11.2 Department Summary Report

For every department, display:

- Department name
 - Employee count
 - Active employee count
 - Total salary
 - Average salary
 - Highest-paid employee
 - Average performance rating
-

11.3 Promotion Eligibility Report

The report should contain:

- Employee name
- Department

- Joining date
 - Experience
 - Performance rating
 - Eligibility status
 - Reason for non-eligibility
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11.4 Bonus Report

The report should contain:

- Employee ID
 - Employee name
 - Salary
 - Rating
 - Bonus percentage
 - Bonus amount
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11.5 Payroll Report

The payroll report should contain:

- Total active employees
 - Total basic salary
 - Total bonus
 - Total payable payroll
 - Department-wise payroll amount
-

12. Assignment Tasks

Learners should complete the following tasks:

1. Find employees whose salary is above the company average.
2. Find the second-highest-paid employee.
3. Find the department with the highest payroll cost.
4. Find the department with the highest employee count.
5. Find employees who joined after 2022.
6. Find employees whose names start with a particular letter.
7. Find employees working under a particular manager.
8. Find employees located in a particular city.
9. Generate a comma-separated employee name list.
10. Find employees having duplicate names.
11. Find the highest-rated employee in each department.

12. Calculate total bonus department-wise.
 13. Calculate the average experience of employees.
 14. Find employees with the same joining year.
 15. Generate separate reports for every department asynchronously.
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13. Advanced Assignment

Develop an employee analytics dashboard summary containing:

- Total employees
- Active employees
- Inactive employees
- Total salary expense
- Average salary
- Highest-paid employee
- Lowest-paid employee
- Highest-rated employee
- Longest-serving employee
- Recently joined employee
- Promotion-eligible employee count
- Department with highest payroll cost

All calculations should be performed using Java 8 features.

14. Testing Scenarios

The application should be tested using the following scenarios:

Scenario 1: Empty Employee List

Expected result:

- Application should not fail.
- Salary calculations should return a safe default.
- Search operations should return an appropriate message.

Scenario 2: Employee Not Found

Expected result:

- No null pointer exception should occur.
- Optional should handle the missing employee.

Scenario 3: Invalid Salary

Expected result:

- Employee record should be rejected.
- A meaningful validation message should be displayed.

Scenario 4: Invalid Rating

Expected result:

- Ratings below 1 or above 5 should not be accepted.

Scenario 5: Inactive Employee

Expected result:

- Employee should not be included in active payroll.
- Employee should appear in the inactive employee report.

Scenario 6: Same Salary for Multiple Employees

Expected result:

- Sorting should display all matching employees correctly.
- Highest salary operations should handle equal salaries.

Scenario 7: Future Joining Date

Expected result:

- Employee record should be rejected.
- Experience calculation should not be performed.

15. Learning Outcomes

After completing this case study, learners should be able to:

- Explain the purpose of Java 8 features.
- Implement lambda expressions.
- Create custom functional interfaces.
- Use built-in functional interfaces.
- Process collections using Stream API.
- Filter, transform and sort employee records.

- Perform salary aggregation.
- Group and partition employee data.
- Handle missing values using Optional.
- Work with LocalDate and Period.
- Create default and static interface methods.
- Use method references.
- Perform asynchronous operations using CompletableFuture.
- Build a layered Java application using Java 8.
- Apply Java 8 concepts to a realistic business problem.